

Name _____



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination
Higher 2

Class	Reg Number
17	

Calculator Model / No.

Chemistry

9729/01

Paper 1

21 September 2018
1 hour

Additional Material: *Data Booklet*
 OMS

READ THESE INSTRUCTIONS FIRST

Write your name, class and register number in the spaces provided at the top of this page.

There are **thirty** questions in this paper. Answer **all** questions. For each question, there are four possible answers labelled **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the OMR answer sheet.

Read very carefully the instructions on the use of OMR answer sheet.

You are advised to fill in the OMR Answer Sheet as you go along; no additional time will be given for the transfer of answers once the examination has ended.

Use of OMR Answer Sheet

Ensure you have written your name, class register number and class on the OMR Answer Sheet.

Use a **2B** pencil to shade your answers on the OMR sheet; erase any mistakes cleanly. Multiple shaded answers to a question will not be accepted.

For shading of class register number on the **OMR sheet**, please follow the given examples:

If your register number is **1**, then shade **01** in the index number column.

If your register number is **21**, then shade **21** in the index number column.

This document consists of **16** printed pages (including this cover page)

Answer all questions.

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

- 1** Zinc ethanoate, $(\text{CH}_3\text{CO}_2)_2\text{Zn}$ ($M_r = 183.4$) may be taken as a dietary supplement to prevent zinc deficiency.

What is the total number of ions present in a 5 cm^3 solution of aqueous zinc ethanoate given that the solution has a concentration of 10.64 g dm^{-3} ?

- A** 1.7×10^{19} **B** 5.2×10^{20} **C** 3.5×10^{22} **D** 1.0×10^{23}

- 2** Gases given off during volcanic eruptions include H_2S and CS_2 .

A 40 cm^3 gaseous sample of H_2S and CS_2 , present in a 3 : 1 ratio respectively, was analysed by combustion using 100 cm^3 of oxygen. After measuring the volume of gas remaining, the product was treated with an excess of aqueous sodium hydroxide and the volume of gas measured again.

Any sulfur present is converted to SO_2 after combustion.

Given that all volume measurements were made under room conditions, what were the measured volumes?

	volume of gaseous mixture after burning / cm^3	volume of gaseous mixture after adding NaOH (aq) / cm^3
A	60	0
B	60	50
C	85	25
D	85	75

- 3** Consider the following reactions:



Which of the following correctly shows the increasing order of reducing strength of the three species, Cl_2 , F_2 and NH_3 ?

- A** $\text{F}_2 < \text{Cl}_2 < \text{NH}_3$ **B** $\text{Cl}_2 < \text{F}_2 < \text{NH}_3$
C $\text{NH}_3 < \text{F}_2 < \text{Cl}_2$ **D** $\text{NH}_3 < \text{Cl}_2 < \text{F}_2$

- 4 The successive ionization energies (I.E.) of two elements, **L** and **M**, are shown below:

IE / kJ mol ⁻¹	1 st	2 nd	3 rd	4 th	5 th	6 th	7 th	8 th
L	1000	2252	3357	4556	7004	8496	27107	31719
M	578	1817	2745	11577	14842	18379	23326	27465

What is the likely formula of the compound formed when **L** and **M** reacts together?

- | | | | |
|----------|-----------------------------------|----------|-----------------------------------|
| A | L₃M₂ | B | L₂M₃ |
| C | LM₃ | D | L₃M |

- 5** Gaseous particle X has a charge of +1 and a proton number, n .

Gaseous particle Y has a proton number of $(n+1)$ and is isoelectronic with X.

Which of the following statements correctly describes **X** and **Y**?

- A** X and Y are isotopes.
- B** X and Y atoms have same full electronic configurations.
- C** Y has a charge of +1 and same charge density as that of X.
- D** Y has a charge of +2 and smaller ionic radius than X.

- 6** Which of the following pairs of molecules satisfies both of the following conditions?

- Only one molecule is polar.
- The second molecule has a larger bond angle than the first molecule.

	First molecule	Second molecule
A	CO ₂	F ₂ O
B	NF ₃	SO ₃
C	SF ₆	XeF ₄
D	SO ₂	NO ₂

7 Which of the following statements about Al is correct?

- 1 It has high melting point due to strong attraction between the delocalised electrons and residual cations.
- 2 It has stronger metallic bonding than Na.
- 3 It has high electrical conductivity due to the ability of its ions to carry the current.

A 1, 2 and 3

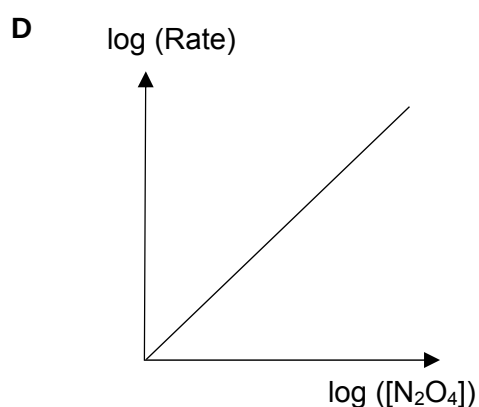
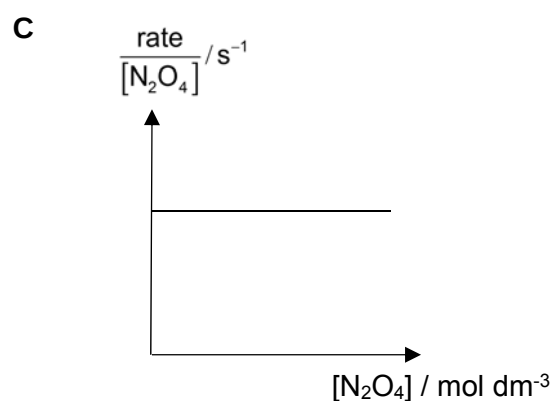
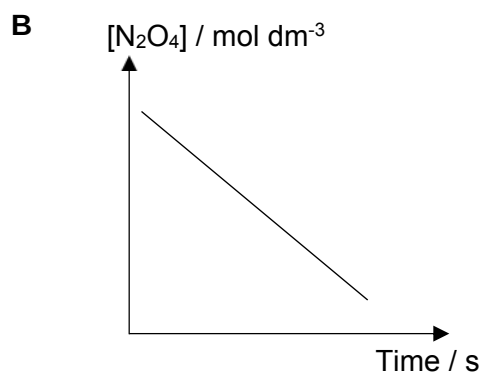
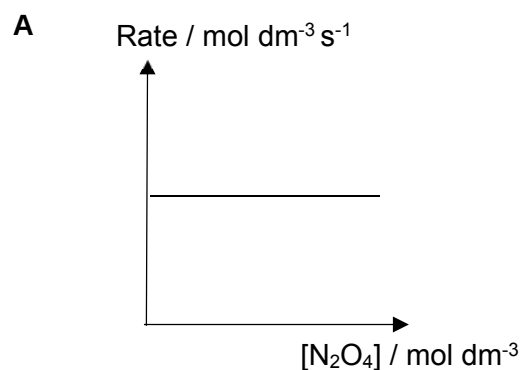
B 1 and 2 only

C 1 only

D 2 and 3 only

8 The decomposition of dinitrogen tetroxide, N_2O_4 is found to be first order with respect to the concentration of N_2O_4 .

Which of the following graphs confirms the above finding?

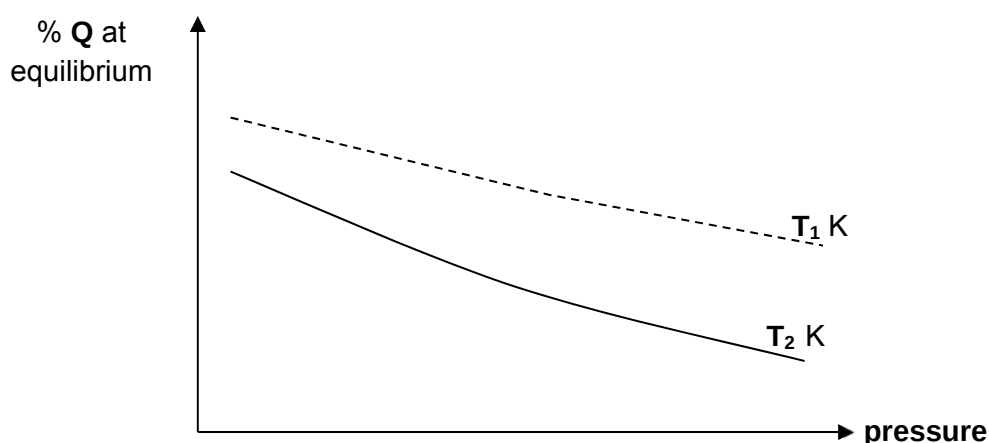
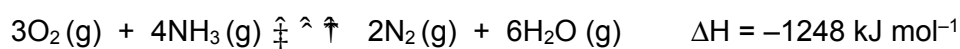


- 9 The hydrolysis of the anticancer drug *cis-platin* in water follows a first-order kinetics with a rate constant of 0.09 h^{-1} at 25°C .

How long will it take for the concentration of a freshly prepared aqueous solution of *cis-platin* to decrease to 18% of its original concentration?

- | | | | |
|----------|--------|----------|--------|
| A | 7.7 h | B | 15.4 h |
| C | 19.0 h | D | 42.8 h |

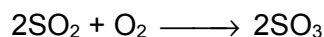
- 10** The graph below shows the variation of the percentage of **Q** present at equilibrium, with temperature and pressure.



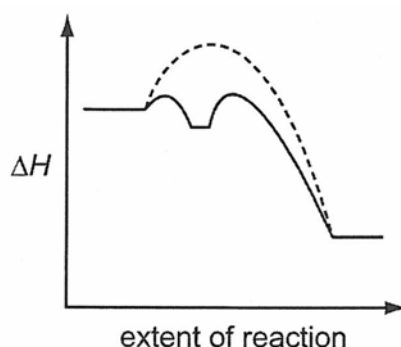
Which of the following systems could **Q** represent?

	Q	Temperature
A	O ₂	T ₁ > T ₂
B	NH ₃	T ₂ > T ₁
C	N ₂	T ₁ > T ₂
D	H ₂ O	T ₂ > T ₁

- 11 The uncatalysed reaction between SO_2 and O_2 is slow.



This reaction is speeded up in the presence of a suitable catalyst. The reaction profile diagram shows the energy changes involved during the catalyzed and uncatalysed reaction.



The uncatalysed reaction is shown as a dashed line.

Which of the following statements is true?

- 1 It involves a homogenous catalyst.
- 2 It involves a stable intermediate.
- 3 Both the activation energies of the forward and backward reaction are lowered in the presence of the catalyst.

- | | |
|-----------------------|---------------------|
| A 1 only | B 3 only |
| C 2 and 3 only | D 1, 2 and 3 |

- 12 Water dissociates into ions according to the following equilibrium.



Which of the following is **incorrect** as temperature is increased to 50°C ?

- A** The dissociation constant of water, K_w , increases
- B** pH of water remains at 7
- C** $[\text{H}^+]$ increases.
- D** $[\text{H}^+] = [\text{OH}^-]$

- 13 Which of the following pair of solutions will form an alkaline buffer solution that **best** resists pH changes when a small amount of acid or base is added.

- A 10 cm³ of 0.1 mol dm⁻³ NaOH and 20 cm³ of 0.2 mol dm⁻³ NH₄⁺Cl⁻
- B 25 cm³ of 0.10 mol dm⁻³ NaOH added to 50 cm³ 0.10 mol dm⁻³ CH₃CO₂⁻Na⁺
- C 10 cm³ of 1 mol dm⁻³ H₂SO₄ and 20 cm³ of 1 mol dm⁻³ CH₃CH₂NH₂
- D 50 cm³ of 0.05 mol dm⁻³ NaOH added to 50 cm³ 0.10 mol dm⁻³ CH₃NH₃⁺Cl⁻

- 14 The table shows some data of two acid–base indicators.

indicator	approximate pH range of colour change	colour change	
		acid	alkali
bromophenol–blue	3.0 – 4.6	yellow	purple
phenol–red	6.8 – 8.5	yellow	red

Which conclusion can be drawn about a solution when it turns bromophenol–blue purple and phenol–red yellow?

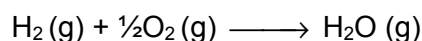
- A The solution is weakly acidic.
- B The solution is strongly acidic.
- C The solution is weakly alkaline.
- D The solution is strongly alkaline.
- 15 Below is one of the propagation steps in the reaction between CH₄ and Cl₂.



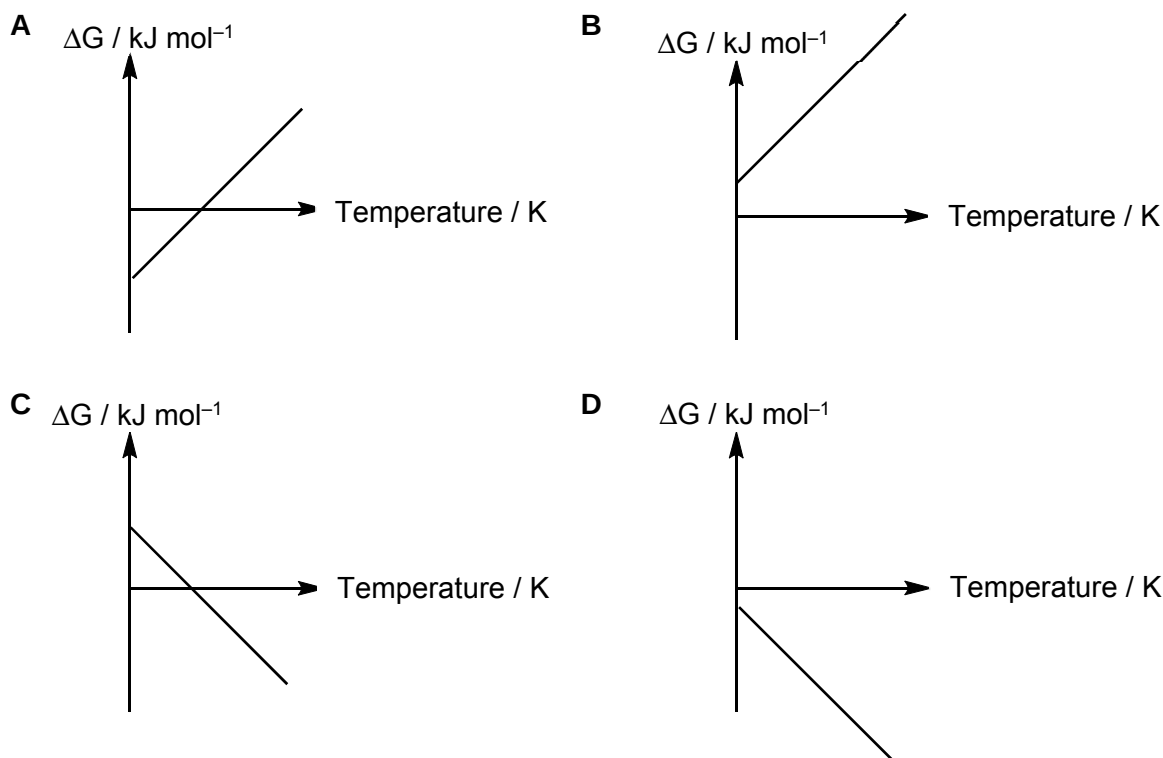
Given the standard enthalpy changes of formation for CH₄, CH₃ radical and HCl are provided, which other enthalpy value is required to calculate $\Delta H^\ominus_r$?

- A Standard enthalpy change of atomisation of chlorine
- B Standard enthalpy change of formation of HCl (aq)
- C Standard enthalpy change of formation of Cl₂ gas
- D Bond dissociation value for C–H bond

- 16 The combustion of gaseous hydrogen is a strongly exothermic process with equation as follows:



Which of the following graphs best illustrates how ΔG varies with temperature?



- 17 A student wanted to predict the solubility of barium chloride using calculation based on the following values.

Enthalpy change	Numerical value / kJ mol^{-1}
Lattice energy of barium chloride	2053
$\Delta H^\circ_{\text{hydration}}$ for Ba^{2+}	1309
$\Delta H^\circ_{\text{hydration}}$ for Cl^-	378

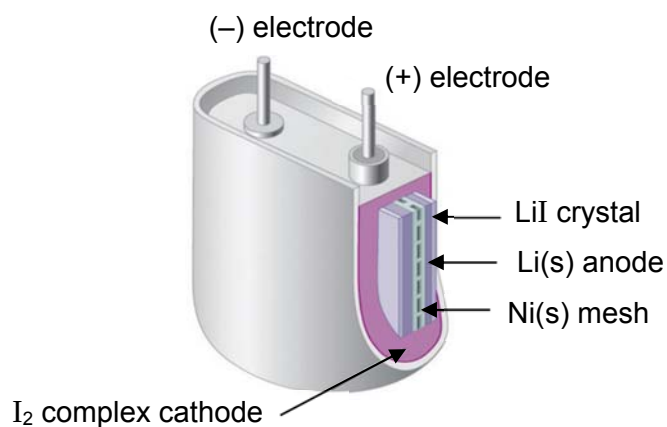
The student wrongly concluded that “barium chloride has $\Delta H^\circ_{\text{solution}}$ of 366 kJ mol^{-1} , hence it is insoluble.”

Which of the following should be the correct conclusion?

- A Barium chloride is insoluble. However, $\Delta H^\circ_{\text{solution}}$ should be $+366 \text{ kJ mol}^{-1}$, the positive sign needs to be shown.
- B Barium chloride is insoluble. However, $\Delta H^\circ_{\text{solution}}$ should be $+12 \text{ kJ mol}^{-1}$.
- C $\Delta H^\circ_{\text{solution}}$ should be -366 kJ mol^{-1} , hence barium chloride is soluble
- D $\Delta H^\circ_{\text{solution}}$ should be -12 kJ mol^{-1} , hence barium chloride is soluble

- 18 Use of the Data Booklet is relevant for this question.

Artificial pace makers are used to regulate the heartbeat of cardiac patients. The device is powered by a water-free lithium iodide battery as shown below.

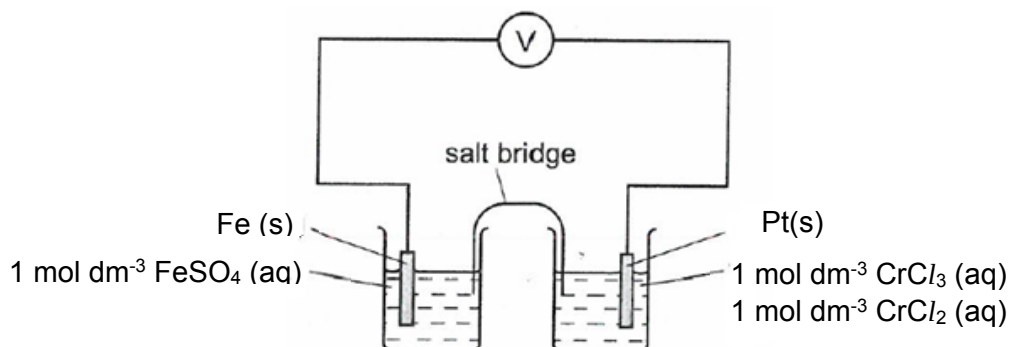


Which of the following statements about the lithium iodide battery is correct?

- A The I₂ complex cathode carries a negative charge.
- B The cell potential of the lithium iodide battery has a value of +2.50 V.
- C The Ni mesh provides a medium for Li⁺ ions to flow from Li (s) to I₂ complex.
- D The pacemaker will last for less than 3 years if it contains 5 g of reactive Li(s) and operates at 0.8 mA.

- 19 Use of the Data Booklet is relevant for this question.

The following electrochemical cell was set up.



Which change to the half-cells could result in a cell potential of 0.00 V?

- 1 increase $[\text{FeSO}_4]$
- 2 decrease $[\text{CrCl}_2]$
- 3 decrease $[\text{CrCl}_3]$
- 4 increase the surface area of the Fe electrode

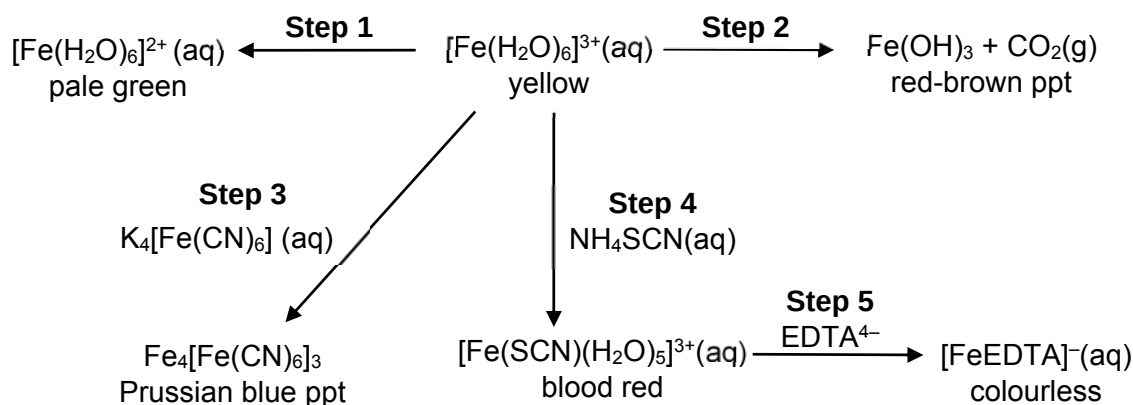
A 1 and 2 only

B 1 and 3 only

C 1, 2 and 3 only

D 3 and 4 only

- 20 The following shows a reaction scheme involving aqueous Fe(III) ions.



Which of the following statements regarding the reaction scheme shown above is correct?

- A Step 1 and Step 4 are ligand exchange reactions.
- B The increasing order of ligand strength is $\text{H}_2\text{O} < \text{CN}^- < \text{SCN}^- < \text{EDTA}^{4-}$.
- C $\text{NaOH}(\text{aq})$ can be used as a reagent in Step 2.
- D Both Fe(II) and Fe(III) ions exist in the $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$.

21 Use of the Data Booklet is relevant for this question.

Vanadyl (IV) sulfate, $\text{VO}(\text{SO}_4)(\text{H}_2\text{O})_5$, is a hygroscopic blue solid and is a common source of vanadium in the laboratory.

Which of the following statements is true for an aqueous solution of $\text{VO}(\text{SO}_4)$?

- 1** $\text{VO}^{2+}(\text{aq})$ can serve as a homogeneous catalyst in the reaction between iodide ions and peroxodisulfate ions.
- 2** The solution appears blue as the d-orbitals are split by the presence of the sulfate ions.
- 3** Zn metal can reduce $\text{VO}^{2+}(\text{aq})$ to $\text{V}^{2+}(\text{aq})$.

A 1 and 2 only

B 1 and 3 only

C 2 and 3 only

D 1, 2 and 3

22 The properties of the oxides of four elements in Period 3, **E**, **F**, **G** and **H** are given below.

- 1** The oxide of **E** is amphoteric.
- 2** The oxide of **F** dissolves in water to form a strongly alkaline solution.
- 3** The oxide of **G** reacts with dilute sodium hydroxide at room temperature.
- 4** The oxide of **H** is insoluble in water but is soluble in concentrated sodium hydroxide.

Which of the following shows the correct sequence of the four elements in order of increasing proton number?

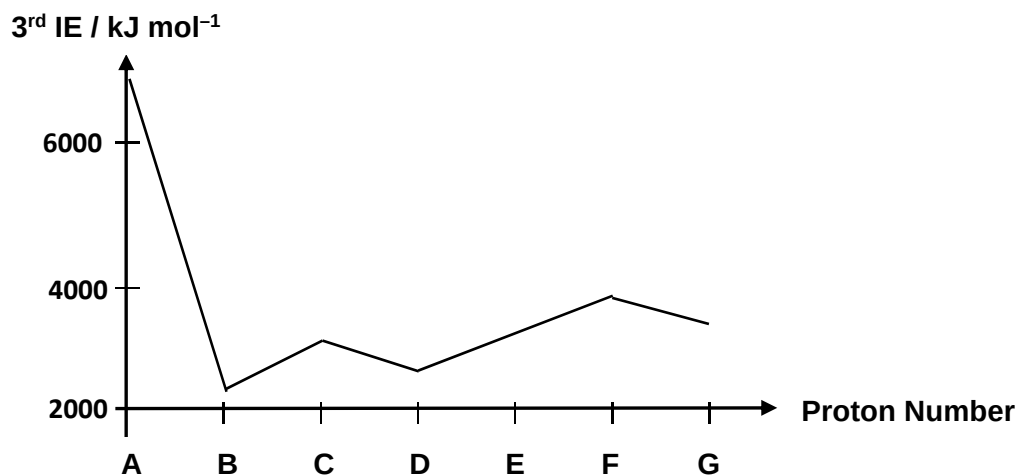
A F, E, G, H

B F, E, H, G

C G, F, E, H

D G, F, H, E

- 23 The diagram shows the third ionisation energy of seven consecutive elements, **A** to **G** in the Periodic Table. Their atomic numbers lie between 3 and 20.



Which of the following statements is correct?

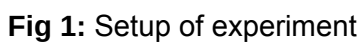
- 1 Element **A** is strongly oxidising.
- 2 Oxide of **B** is soluble in water.
- 3 Oxide of **C** is acidic.
- 4 Element **E** forms a hydride of general formula H_2E .

A 1 and 4 only

B 1, 3 and 4 only

C 2, 3 and 4 only

D 3 and 4 only



<i>Carbonate</i>	<i>Colour before heating</i>	<i>Colour after heating</i>	<i>Observations</i>
CaCO ₃	white	white	white ppt formed with limewater
BaCO ₃	white	white	white ppt formed with limewater
CuCO ₃	green	black	white ppt formed with limewater

Using the above data and relevant data from the *Data Booklet*, which of the following statements concerning the metal carbonates is correct?

- 1 Lattice energy of calcium carbonate is more exothermic than that of barium carbonate.
- 2 Carbon dioxide is produced at a lower temperature from calcium carbonate than from barium carbonate.
- 3 The oxide from the copper carbonate is formed more readily than that from calcium carbonate.

- A** **1, 2 and 3**
- B** **1 and 2 only**
- C** **1 and 3 only**
- D** **2 only**

- 25** 3-ethylpentane can react with bromine in the presence of sunlight to give a mixture of three possible monosubstituted halogenoalkanes, 1-bromo-3-ethylpentane, 2-bromo-3-ethylpentane and 3-bromo-3-ethylpentane.

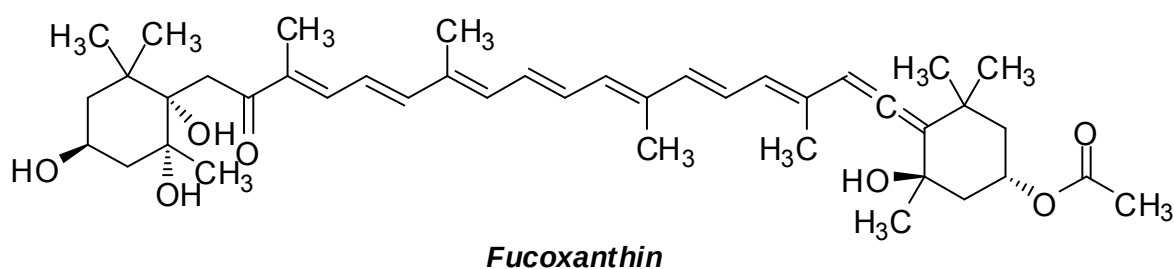
Given the relative rates of abstracting H atoms are:

Type of H atom	Primary	Secondary	Tertiary
Relative rate of abstraction	1	4	6

What is the expected ratio of 1-bromo-3-ethylpentane, 2-bromo-3-ethylpentane and 3-bromo-3-ethylpentane formed?

- | | | | |
|----------|-----------|----------|-----------|
| A | 1 : 1 : 1 | B | 1 : 4 : 6 |
| C | 3 : 8 : 2 | D | 9 : 6 : 1 |

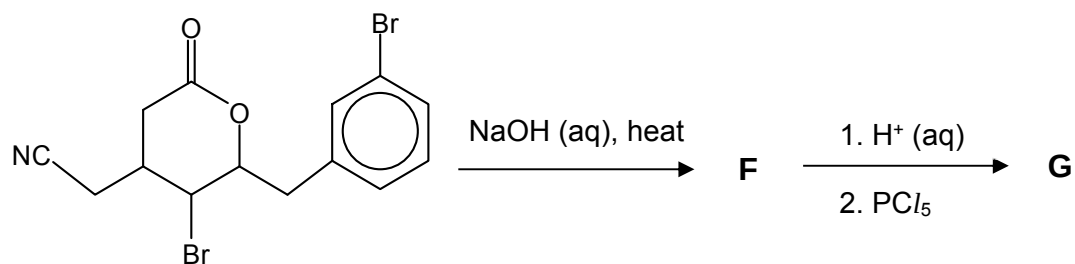
- 26** *Fucoxanthin* is a carotenoid which is found as an accessory pigment in the chloroplasts of brown algae and most other heterokonts, giving them a brown or olive-green colour.



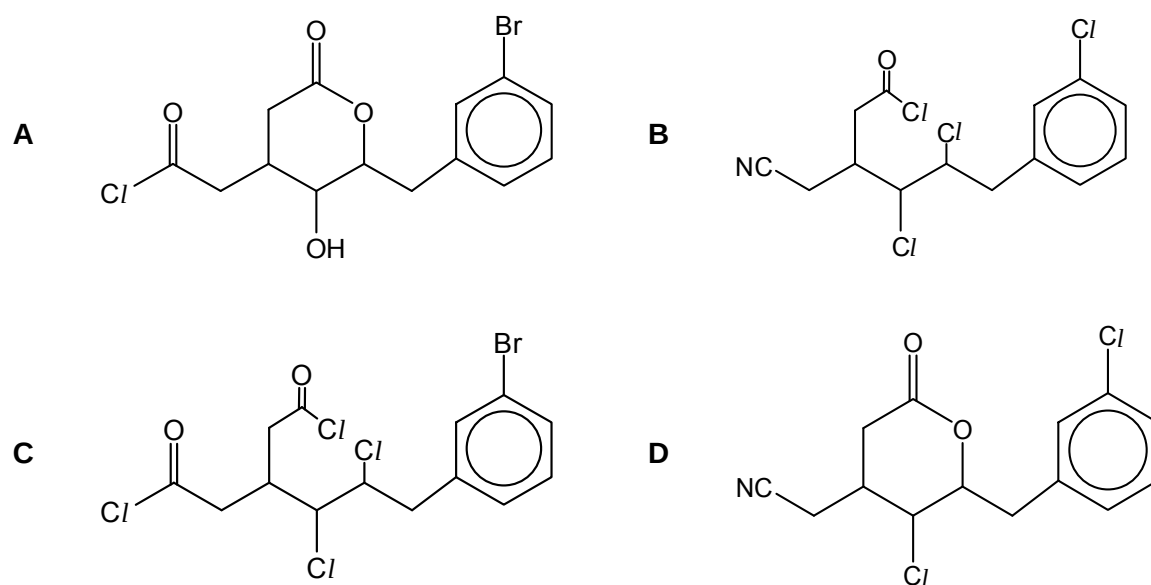
How many organic compounds will be formed when *Fucoxanthin* reacts with hot acidified KMnO_4 ?

- | | | | |
|----------|---|----------|---|
| A | 3 | B | 4 |
| C | 5 | D | 6 |

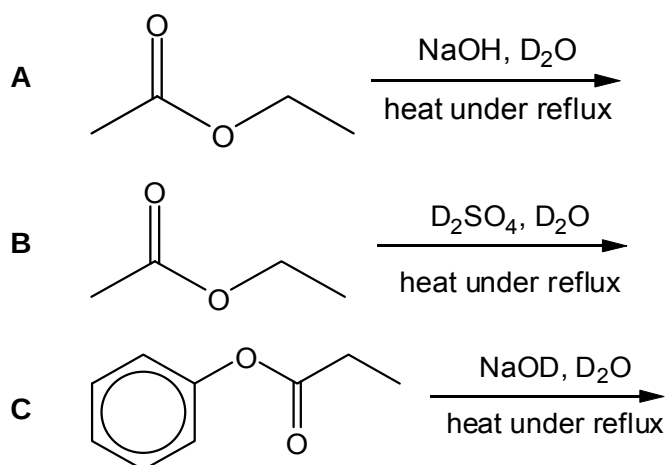
- 27 The reaction scheme below shows the formation of compound **G**:

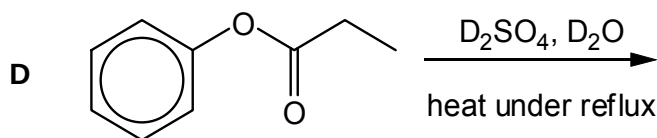


What is the structure of compound **G**?

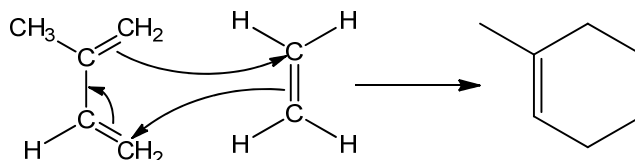


- 28 Which of the following reactions will **not** incorporate deuterium ($\text{D} = {}^2_1\text{H}$, an isotope of hydrogen) into any of the organic products formed? ($\text{D} = {}^2_1\text{H}$, an isotope of hydrogen)

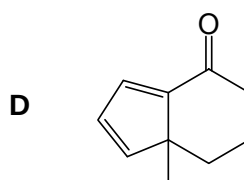
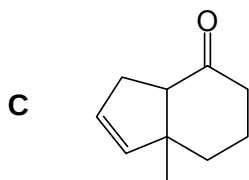
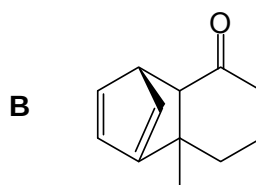
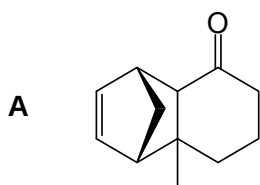
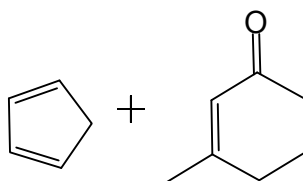




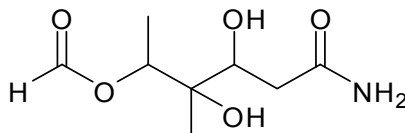
- 29** *Limonene* can be obtained via a series of organic reactions involving the *Diels–Alder* reaction. The *Diels–Alder* reaction involves a concerted cycloaddition mechanism.



What would be the product form when the following diene and alkene reacts in a 1:1 ratio?



- 30** Which of the following statements is true of compound **A**, shown below?



Compound **A**

- 1 Upon reaction with hot acidified KMnO_4 , an organic compound containing 5 oxygen atoms is formed.
- 2 Compound **A** has a total of 6 stereoisomers.
- 3 On reacting with LiAlH_4 , one molecule with one $-\text{OH}$ group is formed together with another molecule that has 4 $-\text{OH}$ groups.

A 1 and 2 only

B 2 and 3 only

C 1 only

D 3 only

END OF PAPER